SPEEDUINO 0.4 ECU PINOUT			
35 PIN - AMP CONNECTOR			
PIN	FUNCTION	COMMENTS	NOTES
1	INJECTOR - 1		
2	INJECTOR - 2		
3	INJECTOR - 3		
4	INJECTOR - 4		
5	IGNITION - 1		
6	IGNITION - 4		
7	IGNITION - 3		
8	IGNITION - 2		
9	BOOST OUT		
10	VVT		
11	IDLE - 2		
12	PWM IDLE		
13	FLEX FUEL		
14	RADIATOR FAN - OUT		
15	CLUTCH SWITCH		
16	IAT SENSOR		
17	WB02 SENSOR		
18	CRANK IN		
19	VR1 -		
20	5V - OUT		
21	STEPPER - A1		
22	STEPPER - A2		
23	ECU GND - IN		
24	GND - OUT		
25	EXT. MAP		
26	FUEL PUMP - OUT		
27	TACHO - OUT		
28	CLT SENSOR		
29	TPS SENSOR		
30	CAM - IN		
31	VR2 -		
32	GND - OUT		
33	STEPPER - B2		
34	STEPPER - B1		
35	12V - ECU LIVE - IN		

12 PIN - CONNECTOR			
PIN	FUNCTION	COMMENTS	NOTES
1	CAN - H		
2	LAMP - 5/SPARE INPUT		SOLDER JUMPER FOR LAMP OUTPUT
3	LAMP - 6/SPARE INPUT		SOLDER JUMPER FOR LAMP OUTPUT
4	LAMP - 7/SPARE INPUT		SOLDER JUMPER FOR LAMP OUTPUT
5	LAMP - 8/SPARE INPUT		SOLDER JUMPER FOR LAMP OUTPUT
6	SPARE INPUT		
7	CAN - L		
8	LAMP - 4		
9	LAMP - 3		
10	LAMP - 2		
11	LAMP - 1		
12	SPARE INPUT		

Notes:

Lamp outputs ideal for Check engine light, Rpm limits, Launch control, Temperature warnings etc.

Spare inputs (J7,J8) ideal for features such as VSS, Oil pressure and other optional inputs.

CAN LOW/HIGH - Canbus output for digital clusters and gauges.

Solder TX & RX To Mega Serial 3.

Solder J10 - 12V And GND To Speeduino Voltage Inputs.

If suffering with voltage issues onboard the PCB. Please do the following -

Solder wire jumpers across - 5VA Main - To 5VB & 5VC And so on. (J1,J2,J3)

Solder wire jumpers across - GND1A Main - To GND1B & GND1C And so on. (J4,J5,J6) DO NOT CONNECT TO J10 GND. THIS IS ECU INPUT (BODY/CHASSIS).

12 PIN - CONNECTOR				
PRODUCT	QUANTITY	PLACEMENT	NOTES	LINK
ESP32 WROOM32 DEVKIT C 30PIN	X1	U2		https://www.aliexpress.com/item/1005006476900011.html
MCP2515 CAN MODULE	X1	U1		https://www.aliexpress.com/item/1005006646252397.html
JAE MX23A12NF1	X1	CN2	12-PIN CONNECTOR	https://www.aliexpress.com/item/1005010178343640.html
IDC HEADER 2x20 P2.54MM FEMALE	X1	CN3	FEMALE HEADER TO SPEEDUINO	https://www.aliexpress.com/item/1005008966856219.html
TYCO AMPSEAL 776164-1 ANGLE	X1	CN1	35-PIN CONNECTOR	https://www.aliexpress.com/item/1005003107028993.html
AO3400A SOT-23 MOSFETS	X8	Q1 - TO - Q8	FOR LAMP LOGIC	https://www.aliexpress.com/item/1005009307642127.html
0603 10K 4X RESISTOR ARRAY	X2	RN3,RN4	NETWORK ARRAY	https://www.aliexpress.com/item/1005007204291786.html
0603 1K 4X RESISTOR ARRAY	X2	RN1,RN2	NETWORK ARRAY	https://www.aliexpress.com/item/1005007204291786.html
0603 10K RESISTOR'S	X2	R1,R4		SAME LINK AS BELOW
0603 20K RESISTOR'S	X2	R2,R5		https://www.aliexpress.com/item/1005002991861666.html
0603 47K RESISTOR'S	X2	R3,R6		SAME LINK AS ABOVE
MALE 2.54 HEADERS		CUT TO SIZE		https://www.aliexpress.com/item/1005009370767269.html